The Cookie Circle

Imaginary Numbers and Euler's Equation

November 4, 2026 · Nathaniel

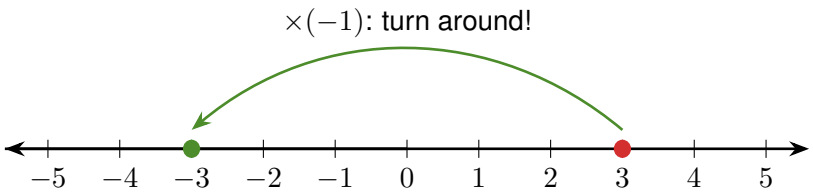
Name: _____ Date: _____

The story

Elmo loves to dance on the number line! Elmo stands on a number and faces the way the numbers grow. Elmo knows two dance moves: **walking** and **turning**.

Cookie Monster has baked one GIANT cookie, and **The Count** has a very tricky riddle. To solve it, Elmo will discover a whole new kind of number, and one of the most famous equations in all of math!

Oscar's opposite trick: times -1 means "turn around"



Oscar the Grouch loves opposites! His favorite trick is multiplying by -1. It is a **half turn**: Elmo spins around to face the other way, and the number jumps to the other side of 0. So $3 \times (-1) = -3$. *"Scram! Go the other way!"*

Question 1. $4 \times (-1) =$ _____

Question 2. Turn around *two* times: $2 \times (-1) \times (-1) =$ _____

Which way is Elmo facing now? _____

Try it! Stand up and face the board. Do a half turn. Do another half turn. Which way are you facing?

The tricky riddle

The Count asks: *"Is there a number that, if you multiply by it **two** times, makes Elmo turn around? Ah ah ah!"*

Question 3. Elmo tries. Fill in the blanks.

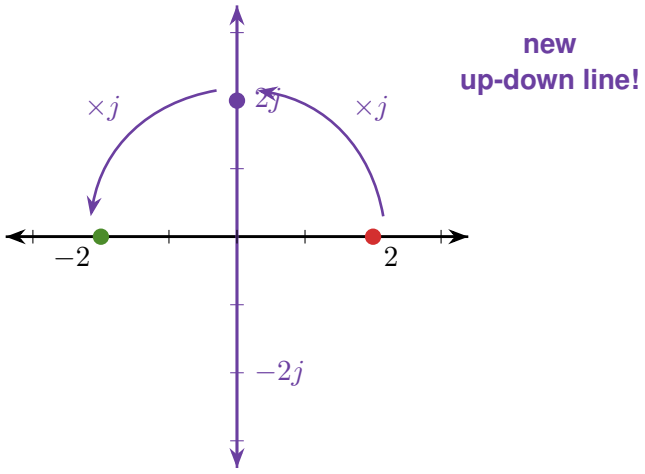
$1 \times 1 =$ _____ Did Elmo turn around? _____

$(-1) \times (-1) =$ _____ Did Elmo turn around? _____

Nothing on the number line works! Elmo needs a brand new idea.

Elmo's quarter turn, j

Try it! Do a half turn using **two quarter turns**: turn left a little (a quarter turn), then turn left again. You are facing backwards. It worked!



People invented a new number for this. Mathematicians call it i, but **engineers**, who build radios and phones, call it j (they already use i for electricity). We are engineers today, so we write j! Multiplying by j is a **quarter turn** to the left. It makes Elmo dance off the number line onto a new **up-down line**. Numbers on the up-down line are called **imaginary numbers**. Two quarter turns make a half turn, so $j \times j = -1$. Riddle solved!

The turning pattern

Elmo starts at 1 and keeps multiplying by j. Each time, Elmo does one more quarter turn, and The Count counts them: *"ONE quarter turn! TWO quarter turns! Ah ah ah!"* Use the picture on the page before to fill in where Elmo lands.

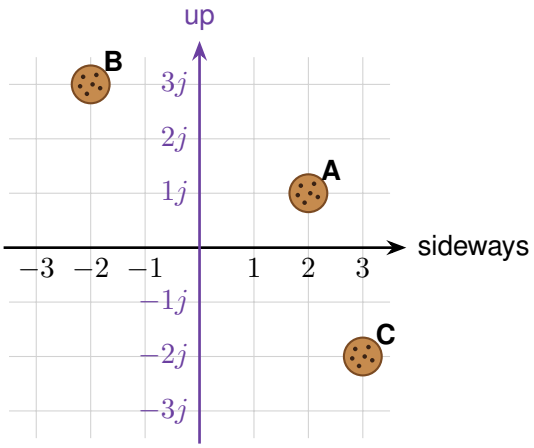
Quarter turns	Elmo is at	Pointing
0	1	right
1	j	up
2		
3		
4		

Question 4. How many quarter turns bring Elmo all the way back to 1? _____

Question 5. The pattern repeats! Where is Elmo after 8 quarter turns? (Count by 4s.) _____

Cookie Monster's cookie map

Now numbers can go **sideways** and **up**! Cookie Monster hid cookies all over the map. A number like $2 + 1j$ is an address: walk 2 steps right, then 1 step up. *"Me want cookie at that address!"*



Question 6. Cookie **A** is at $2 + 1j$. Write the addresses of cookies B and C.

B: _____ + _____ j **C**: _____ + _____ j

Question 7. Cookie Monster hid one more cookie at $1 + 3j$. Draw it on the map!

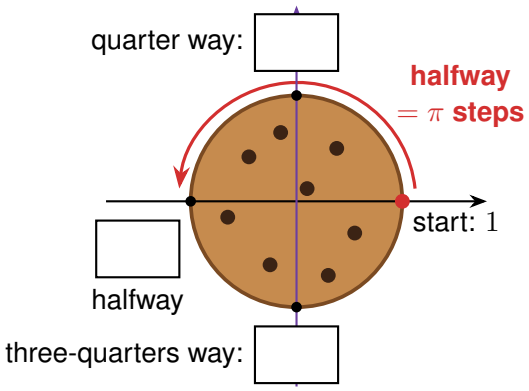
Elmo walks around the giant cookie

Try it! Wrap a string once around a cup (or a real cookie!). Then lay the string flat and count how many cup-widths fit along it, like The Count: *"ONE! TWO! THREE! And a little bit more! Ah ah ah!"*

Question 8. About how many cup-widths long was the string? _____

It is always a little more than 3, for *every* circle! That number is called **pi**, written π. It is about 3.14.
If a cookie is 2 steps wide, the edge all the way around is 2π steps long, so **halfway around is exactly π steps**.

Elmo starts at 1 and walks around the edge of Cookie Monster's giant cookie, which is 2 steps wide. Fill in the boxes with where Elmo lands: 1, j, -1, or $-j$.



Euler's equation

How a scientist writes "walk around the cookie"

A mathematician named **Leonhard Euler** (say "OY-ler") found a way to write Elmo's walk around a circle. It uses the special number e, about 2.72, from Cookie Monster's Growing Cookie Jar. When you write

$$e^{j \times \text{steps}}$$

it means: **start at 1 and walk that many steps around the edge of the cookie.**

Question 9. Walk π steps (halfway around). Where does Elmo land?

$$e^{j\pi} = _____$$

Question 10. Now add 1 to both sides. $-1 + 1 =$ _____. So:

$$e^{j\pi} + 1 = _____$$

The Count's five famous numbers

$$e^{j\pi} + 1 = 0$$

"FIVE famous numbers in ONE equation! Ah ah ah!" Many mathematicians say this is the **most beautiful equation** in all of math, because it uses five famous numbers at once.

Question 11. Draw a line from each number to what it means.

0 •	• The quarter-turn number
1 •	• Nothing at all
j •	• The special number for Euler's cookie walk, about 2.72
π •	• Where Elmo starts
e •	• How far halfway around the cookie is, about 3.14

Draw it!

Draw Elmo walking halfway around Cookie Monster's giant cookie, from 1 to -1. Write $e^{j\pi} = -1$ next to your picture.

For teachers: this worksheet uses the engineering convention $j = \sqrt{-1}$ (in electrical engineering i denotes current); mathematics texts write i, so $e^{j\pi} + 1 = 0$ is the same as $e^{i\pi} + 1 = 0$. Multiplying by j rotates the complex plane by $90°$ counterclockwise, which is why $j^2 = -1$ reads as two quarter turns making a half turn. Euler's formula $e^{j\theta} = \cos\theta + j\sin\theta$ places $e^{j\theta}$ on the unit circle at arc length θ from 1 (counterclockwise), so $\theta = \pi$ lands on -1. The "cookie 2 steps wide" is the unit circle (radius 1), whose circumference is 2π. Why e in particular is the base that turns "growth" into "rotation" is a calculus fact ($\frac{d}{d\theta}e^{j\theta} = j\,e^{j\theta}$: the velocity is always a quarter turn from the position) and is beyond this worksheet. Sesame Street characters appear by name only, for classroom use; this worksheet is not affiliated with or endorsed by Sesame Workshop.

Answer key

Question	Answer
1	-4
2	2; facing forward again (the way the numbers grow)
3	$1 \times 1 = 1$, no; $(-1) \times (-1) = 1$, no
Turning table	2 turns: -1, left 3 turns: $-j$, down 4 turns: 1, right
4	4 quarter turns
5	1 (8 is two groups of 4)
6	B is $-2 + 3j$; C is $3 + (-2)j$, which is also written $3 - 2j$
7	A cookie one step right, three steps up
8	A little more than 3 (about 3.14 if measured carefully)
Circle boxes	Quarter way: j; halfway: -1; three-quarters: $-j$
9	-1
10	0; so $e^{j\pi} + 1 = 0$
11	0: nothing at all; 1: where Elmo starts; j: the quarter-turn number; π: halfway around, about 3.14; e: Euler's special number, about 2.72